SANDIP M. PATEL

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PROFESSIONAL SUMMARY

- Lead Embedded & Edge Systems Architect with 16+ years of experience designing and delivering high-performance, real-time systems across Edge, Embedded, and Cloud-integrated platforms.
- Deep expertise in product architecture, edge analytics, and firmware design - spanning multithreaded systems, networking stacks, and real-time data processing for large-scale production environments.
- Proven track record of building end-to-end solutions **(Device → Edge → Cloud)**, with hands-on experience in C/C++, Python, Linux/RTOS, and modern data pipelines **(MQTT, Kafka, S3/MinIO)** for analytics and **ML workflows**.
- Strong background in system design, performance optimization, and hardware–software co-design, with experience spanning network analytics, post-silicon validation systems, and industrial automation products.
- **AWS Certified Solutions Architect – Associate, AWS Certified AI Practitioner, and Lean Six Sigma certified**, bringing a balance of engineering depth and process excellence.
- Experienced in technical leadership, architecture governance, and stakeholder engagement, including client interactions, solution demos, and release management across global teams

Technical Skills:

Languages	C, C++, Python, Shell Script, GO
Operating System	Linux (Embedded & Server), FreeRTOS
Networking & Protocol	TCP/IP, MQTT, Kafka, SNMP, MODBUS, PROFINET, Other L2 OT
Low-Level Interfaces	I2C, SPI, UART/USART (RS232/RS485), 1-Wire
Cloud & Containerization	AWS (EC2, S3, RDS, ECR), MinIO, Docker, Kubernetes
Databases	MySQL, ClickHouse - data debugging, retention/TTL management for metadata & log tables
Hardware Platforms	ARM, STM32, Renesas, ESP32, Raspberry Pi, BeagleBone, Xilinx (FPGA + ARM SoC), Dell Servers
Observability	Grafana
Version Control	Git (GitHub, GitLab, Bitbucket), SVN
IDE& Toolchains	VS Code, GCC, Keil, MPLAB, AVR Studio, CubeSuite
PCB & Design Tools	Altium, Eagle, Proteus
Project Management	JIRA, Confluence

PROFESSIONAL EXPERIENCE

INCEDO INC

Technical Architect — **Oct 2023 – Present**

Sr. Technical Lead — **Aug 2021 – Sep 2024**

- Led design and implementation of the Edge analytics layer within a high-throughput network analytics platform, enabling real-time threat detection via a hybrid streaming + batch (ETL) architecture.
- Engineered security-hardened C/C++ build pipeline aligned with **IEC 62443** practices – static analysis (cppcheck) in CI, pinned-version Docker builds, **Black Duck SCA**/vulnerability management, and **distroless (shell-less)** runtime images to minimize attack surface
- Architected a hybrid Edge–Cloud pipeline (MQTT → Kafka → DB) for real-time edge analysis and cloud ingestion; owned Edge implementation and contributed to cloud data flow design.
- Provided informal technical mentorship to 4+ engineers on embedded/firmware design and secure coding practices; **contributed to fixed-bid proposal scoping and effort estimation**.
- Owned end-to-end release lifecycle (QA → sbx → Prod) and led client-facing discussions, demos & stakeholder engagement.
- Architected concurrent, resource-constrained firmware systems - including low-latency packet processing (libpcap), a DPDK kernel-bypass PoC, and dynamic certificate-based MQTT authentication - prioritizing crash resilience, reduced attack surface
- Optimized containerization (Docker), achieving 78% image size reduction, and supported on-prem production deployments (Dell servers), improving deployment reliability across Kubernetes and edge environments.
- Managed on-prem Kubernetes deployments – authored service YAML manifests, debugged reverse proxy and internal service communication issues, and configured cert termination through to EMQX broker.
- **Adopted agentic AI-assisted development (Kiro)** in client project delivery, reducing a 3-month planned delivery timeline by ~33% while maintaining full SME review; separately led a self-driven AI agent workflow initiative (aligned with company OKRs) to automate recurring tasks and other use cases.

ARASTU SYSTEMS **May 2020 – July 2021**
Technical Lead

- Led end-to-end delivery of the first production lot for a diagnostic board used to validate the functionality of a post-silicon validation platform (**Xilinx FPGA + ARM architecture**) — from **board bring-up** through production release — with custom IP/logic on FPGA controlled via ARM, validating interfaces including SPI, UART, USB, and I2C across multiple configurations.
- Architected a modular, orchestrator-driven validation framework — dynamically launching interface-specific test binaries per upstream command, executing exhaustive configuration sweeps across multiple protocols/interfaces, with dynamic kernel module loading for mutually exclusive interface modes.
- Owned hardware–software co-design and system integration, including porting and customizing embedded Linux on FPGA-based boards, to ensure reliable communication between FPGA fabric and ARM-based software control layers.
- Managed international client engagements — technical reporting, demos & stakeholder communication across geographies.

PIRAMAL WATER PVT. LTD.
R&D Project Manager **April 2018 – April 2020**
Sr. Embedded Design Engineer **April 2016 – March 2018**
Embedded Design Engineer **April 2013 – March 2016**

- Led end-to-end product development - from concept, feasibility analysis, and firmware design through execution and field deployment - translating customer and industry needs into new products aligned with business goals.
- Owned project scope and timeline management across overlapping product initiatives, coordinating cross-functionally with Business Development, Quality, Marketing, and Operations.
- Assessed and validated contract manufacturers and production vendors, including cost analysis and pilot production lot validation for scaled manufacturing readiness.
- Designed firmware architecture and implemented a custom scheduler on bare-metal C (alongside FreeRTOS-based development); maintained **ISO 9001:2015** documentation, clearing audits for 3–4 consecutive years.
- Owned hardware design contributions — component selection, schematic design, and PCB layout — verifying and finalizing designs in close collaboration with hardware engineering, bridging firmware and hardware development end-to-end.
- Resolved on-site installation and field issues through direct customer engagement — hands-on troubleshooting across hardware, firmware, and system-level integration in live deployment environments.

AUTOSOFT SOLUTIONS **Oct 2012 – March 2013**
Jr. Embedded Design Engineer

- Researched and evaluated new technologies across multiple microcontroller platforms.
- Developed embedded system software using compilers, assemblers, and debuggers; wrote portable, optimized code and resolved bugs in existing products.

RESONENT TECHNOLABS PVT. LTD. **April 2011 - Sep 2012**
Embedded Design Engineer

- Conducted requirement analysis and client interactions to define and finalize business requirements; prepared proof-of-concept demos and presentations.
- Developed, tested, and implemented embedded applications.

MAHAVIR SYNTHESIS PVT. LTD **Nov 2009 – March 2011**
Instrument Maintenance Engineer

- Handled installation, troubleshooting, repair, and calibration of plant instrumentation equipment.
- Responded promptly to equipment and machinery breakdowns, keeping the production manager informed on progress.

EDUCATIONAL CREDENTIALS & CERITICATIONS

AWS Certified AI Practitioner, Sep-2025

AWS Certified Solution Architect – Associate, Oct-2024

LEAN SIX SIGMA – White Belt, Mar-2019

MBA (IT), Jan-2019 - ICFAI University

Certificate Course in Embedded System Design, Sep-2009
National Institute of Communication and Information Technology, Pune University, Gandhinagar, 69%

B.E (Electronics & Communication), May-2009
Govt. Engineering College Dahod, Gujarat University, 55%